Anika James

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Summary

PhD in Molecular, Cellular, and Developmental Biology from the University of Kansas with extensive hands-on experience in cellular and molecular biology techniques, including in vivo mouse model studies, mammalian cell culture, transfection, immunoblotting, molecular cloning, protein purification, and recombinant DNA approaches. Skilled in designing and conducting independent research, analyzing and interpreting experimental data, and preparing findings for publication and presentation. Proven track record in laboratory management, mentoring trainees, scientific writing, including grant proposals, and ensuring compliance with institutional biosafety and regulatory standards.

Education

- **PhD in Molecular, Cellular, and Developmental Biology** — Aug 2020 – Apr 2026
  The University of Kansas — GPA: 3.85/4.00

- **B.S. in Biology, Minor in Chemistry** — Aug 2016 – May 2020
  Metropolitan State University of Denver (MSU Denver) — GPA: 3.80/4.00

Research Experience

Graduate Research Assistant — Neufeld Lab, University of Kansas — Aug 2020 – April 2026:

- Designed and conducted hands-on experiments using mammalian cell culture, transfection, recombinant DNA techniques, molecular cloning, and protein-based assays to investigate the tumor suppressor APC in intestinal homeostasis.
- Performed immunoblotting (Western blot), ELISA, immunofluorescence (IF), immunohistochemistry (IHC), and fluorescence in situ hybridization (FISH) to characterize protein expression and localization in cell lines and tissue samples.
- Executed RNA/DNA extraction, PCR, qPCR, and ChIP-PCR to dissect molecular pathways linking APC regulation to inflammation and genome stability.
- Prepared, interpreted, analyzed, and compiled experimental data into peer-reviewed publications, manuscripts, and oral/poster presentations at national conferences.
- Managed daily laboratory operations including supply procurement, inventory management, equipment maintenance, and adherence to safety and biosafety protocols.
- Worked under biosafety protocols for handling biohazardous materials, ensuring compliance with institutional and regulatory standards.
- Mentored and trained undergraduate students and first-year graduate students in experimental design, molecular biology techniques, and data interpretation.
- Contributed to grant proposal writing and applied for available funding mechanisms to support ongoing research.

Undergraduate Researcher — Zajdowicz Lab, MSU Denver — Aug 2019 – May 2020:

- Designed and conducted experiments studying the effects of *Neisseria lactamica* on *Streptococcus mutans* oral biofilm formation using cell culture and molecular assays; presented findings at a national conference.

Undergraduate Researcher — Bonham Lab, MSU Denver — Aug 2018 – Aug 2019:

- Developed electrochemical biosensor approaches involving protein purification and biochemical assays for sensitive serologic detection of pathogenic *Mycoplasma*; reported findings at ASBMB venues.

Teaching & Leadership

Graduate Teaching Assistant — University of Kansas, Lawrence, KS **Spring 2022 – Fall 2023:**

- BIOL 152 (Principles of Organismal Biology) – Spring 2022
- BIOL 688 (The Molecular Biology of Cancer) – Fall 2023

Project Lead, Biology Study Room — University of Kansas, Lawrence, KS **Fall 2021:**

- Directed the comprehensive redesign of BioCenter for Collaborative Learning (BCCL), developing detailed spatial layouts and procurement plans for instructional equipment and collaborative furnishings.

Clinical & Volunteer Experience

Volunteer, Neonatal Intensive Care Unit (NICU) — North Suburban Medical Center, Thornton, CO **Aug 2018 – Jul 2019:**

- Assisted NICU staff with clerical and administrative duties to support efficient unit operations.
- Helped soothe and comfort infants under staff supervision, contributing to a supportive care environment for neonates and their families.

Clinical Shadowing Experience — National Jewish Health / St. Joseph Hospital, Denver, CO **Fall 2019:**

- Shadowed an internal medicine physician, observing patient evaluations, clinical decision-making, and interdisciplinary care in a hospital setting.

Achievements

- **University of Kansas Cancer Center (KUCC) Poster Travel Award** Dec 2025
- **FASEB Trainee Award / Outstanding Poster Presentation Award** Fall 2022
- **Dean's Doctoral Fellowship (Fully Funded)** Fall 2022–Spring 2023; Fall 2020–Spring 2021
- **University of Kansas Cancer Center (KUCC) Poster Travel Award** Dec 2021
- **Certificate of Recognition**, ASBMB 23rd Annual Undergraduate Poster Competition Apr 6, 2019
- **Undergraduate Research Mini-Grants** Nov 12, 2019; Nov 13, 2018
- **Dean and Provost Scholarship** Nov 7, 2019; Apr 5, 2018; Nov 2, 2017
- **CO-WY AMP Research Grant Funding Awards** Nov 7, 2019; Nov 1, 2018

Professional Memberships

- Associate Member, American Association for Cancer Research (AACR) Dec 2024 – Present
- Member, The National Society of Leadership and Success Feb 2019 – Present
- Scientific Writing Scholar, BioKansas & Genetics Society of America 2021 – 2022
- Student Member, American Society for Biochemistry and Molecular Biology (ASBMB) 2018 – 2020

Technical Skills

- **Molecular & Cellular Biology:** Mammalian cell culture, transfection, immunoblotting (Western blot), ELISA, molecular cloning, recombinant DNA techniques, PCR, qPCR, ChIP-PCR, RNA/DNA extraction, protein purification, immunofluorescence (IF), immunohistochemistry (IHC), fluorescence in situ hybridization (FISH)
- **Histology:** H&E staining, Alcian Blue staining, tissue processing, paraffin embedding, microtome sectioning
- **Animal Research:** Mouse colony management (breeding, husbandry), genotyping, ear tagging, teeth trimming, euthanasia, necropsy, tissue/fecal sample collection
- **Imaging & Instrumentation:** Nikon and Keyence microscope operation; equipment calibration, maintenance, and training of lab personnel on instrumentation

- **Data Analysis & Software:** GraphPad Prism, Python, PICRUSt, UCSC Genome Browser, SPSS; MS Office (Word, Excel, PowerPoint); LaTeX
- **Lab Management & Safety:** Supply procurement and inventory management, equipment upkeep, biohazardous waste disposal, compliance with institutional biosafety protocols and regulatory standards for handling biohazardous materials
- **Mentorship & Training:** Training of undergraduate and graduate students in experimental design, techniques, and data interpretation
- **Scientific Writing & Grants:** Authored peer-reviewed manuscripts and grant proposals; BioKansas Scientific Writing Scholar (Genetics Society of America & BioKansas, 2021–2022)

Publications

Peer-Reviewed:

- Sandoval, A. Q., **James, A.**, & Neufeld, K. L. *Nuclear adenomatous polyposis coli elevates STAT1 and reduces CXCL1, 2, and 3 expression and inhibits neutrophil recruitment. Cellular Signaling*, 111957.

Manuscripts submitted:

- **James, A.**, Washington, K., & Neufeld, K. L. *Nuclear APC Deficiency in Apc$^{mNLS/mNLS}$ Mice Unmasks Sexually Dimorphic Colitis Susceptibility and Phenocopies Estrogen Receptor Alpha Loss-of-Function.*

Manuscripts in Progress:

- **James, A.**, Washington, K., & Neufeld, K. L. *Helicobacter hepaticus Colonization Modulates Microbiome and Restores Protection Against Colitis in Apc Wild-Type Mice.* (manuscript in progress)

Presentations

- **Microbiome modulation by *Helicobacter hepaticus* reveals nuclear APC–dependent protection against colitis**
  BIOL 902 – Advanced Molecular Cellular Biology (AMCB) Seminar; Lawrence, KS (Oral) Sep 2025
- **Nuclear APC maintains colon homeostasis and mitigates inflammation**
  Molecular Biosciences Symposium 2025; Lawrence, KS (Poster) Aug 2025
- **Nuclear APC maintains colon homeostasis and mitigates inflammation**
  KUCC Cancer Biology Research Program Meeting; Olathe, KS (Oral & Poster) Jul 2025
- **Nuclear APC maintains colon homeostasis and mitigates inflammation**
  KU Center for Genomics 4th Annual Symposium; Lawrence, KS (Poster) May 2025
- **Nuclear APC maintains colon homeostasis and mitigates inflammation**
  AACR Annual Meeting 2025; Chicago, IL (Poster) Apr 2025
- **Nuclear APC maintains colon homeostasis and mitigates inflammation**
  Biology Senior Seminar, Baker University; Baldwin City, KS (Oral) Mar 2025
- **Nuclear Localization of APC Is Essential for Mucus Barrier Integrity and Mitigation of Colonic Inflammation: A Sex-Specific Study**
  23rd Annual K-INBRE Symposium; Kansas City, MO (Poster) Jan 2025
- **Nuclear Localization of APC Is Essential for Mucus Barrier Integrity and Mitigation of Colonic Inflammation**
  Molecular Biosciences Fall Seminar; Lawrence, KS (Oral) Dec 2024
- **Nuclear Localization of APC Is Essential for Colon Mucus Barrier Integrity and Mitigation of Colonic Inflammation: A Sex-Specific Study**
  KUCC Research Week 2024; Overland Park, KS (Poster) Nov 2024
- **The Role of Nuclear APC in Regulating MUC2 Expression and Colonic Inflammation**
  KUCC Cancer Biology Research Program Meeting; Olathe, KS (Poster) Jul 2024
  KU Center for Genomics 3rd Annual Symposium; Lawrence, KS (Poster) May 2024
  21st Annual K-INBRE Symposium; Overland Park, KS (Poster) Jan 2023
  BIOL 902 AMCB Seminar; Lawrence, KS (Oral) Jan 2024
  KUCC Research Week 2023; Overland Park, KS (Poster) Nov 2023
  FASEB GI Epithelium Conference; Steamboat Springs, CO (Poster) Sep 2023

Molecular Biosciences Symposium; Lawrence, KS (Poster) Aug 2023
KUCC Cancer Biology Research Program Meeting; Olathe, KS (Poster) Aug 2023
KU Center for Genomics 2nd Annual Symposium; Lawrence, KS (Poster) May 2023
BIOL 902 AMCB Seminar; Lawrence, KS (Oral) Feb 2023
21st Annual K-INBRE Symposium; Overland Park, KS (Poster) Jan 2023
KUCC Research Week; Kansas City, KS (Poster) Nov 2022
Molecular Biosciences Symposium; Lawrence, KS (Poster) Aug 2022
FASEB "Gastrointestinal Tract XX"; Steamboat Springs, CO (Poster) Aug 2022

- **The role of nuclear APC in regulating MUC2 expression and colonic inflammation**
  University of Kansas Cancer Center Research Symposium; Kansas City, KS (Oral) Nov 2021

References

- **Dr. Kristi Neufeld**
  Frank B. Tyler Professor of Cancer Research
  Department of Molecular Biosciences, University of Kansas
  Email: `klneuf@ku.edu`
- **Dr. Dan Dixon**
  Professor, Winthrop P. Rockefeller Cancer Institute, University of Arkansas for Medical Sciences
  Email: `ddixon2@uams.edu`
- **Dr. Mizuki Azuma**
  Professor, Department of Molecular Biosciences, University of Kansas
  Email: `azumam@ku.edu`